Maison Rue

A fashion commerce website connecting product discovery, purchasing and after sales service with staff operations and business administration.

Purpose

An introduction to the customer experience, operational portals, technology and data structure.

Scope

Customer storefront · Staff portal · Administrator portal

Edition

7 November 2026 | English

One website for the full shopping journey

Maison Rue presents collections for men, women and children through a restrained fashion storefront. Customers can explore materials and variants, save favourites, place orders and follow service requests.

Discover and choose

Browse seasonal edits, material collections and product categories. Compare colour, size, fit, price and care information before purchasing.

Purchase and follow up

Manage a shopping bag, select delivery details, apply eligible offers and track orders. Delivered items can enter a structured return or exchange process.

Operate and manage

Staff handle orders, inventory and returns. Administrators additionally manage employees, promotions, financial reports and product insights.

Application architecture

The storefront and operational portals share a React application, an Express API and a Microsoft SQL Server database.

Frontend

React 19, React Router 7 and Vite 8 provide page rendering, navigation and development builds. Context providers manage authentication, cart and favourites.

Backend and persistence

Node.js and Express expose APIs for products, accounts, orders, promotions, reporting and returns. SQL Server stores operational records, product variants and transaction history.

Access and integrations

Role checks protect customer, staff and administrator routes. Authentication uses JWT and password hashing with bcrypt. Google address services and the Rue assistant depend on configured external API access.

Browse the collection

The catalogue combines collection navigation with product information and interactive product cards.

Navigation and search

Men, Women, Kids, Accessories, Fall Winter 2026–2027, Cashmere, Linen and Suede collection routes; product search; category and subcategory navigation.

Refine the selection

Filter by price range and colour, switch collection audience and sort by featured order or ascending and descending price.

Product details

Image gallery, colour variants, size selection, price and promotion information, stock checks, quantity controls and wishlist hearts. Expandable details cover fit, care and returns.

Fit assistance

Find my size opens a measurement guide. Size suggestions use available size charts and chest, waist or foot length inputs.

Wishlist and Complete the Look

Two discovery features help customers return to favourite pieces and build coordinated outfits.

Guest wishlist

Visitors can save products without signing in. The wishlist persists in browser localStorage on the same browser and device; clearing site storage removes it.

Account synchronisation

On customer sign in, guest favourites are merged with account favourites. Account favourites are stored through the customer API for later access.

Outfit suggestions

Complete the Look selects complementary garment roles for the same audience and ranks candidates by shared colour palette, material and collection. Suggestions link to product pages.

Current recommendation method

The implementation is rule based. A fallback fills missing suggestion slots from other candidates. Learning from frequently purchased combinations is a future enhancement, not an existing trained recommendation model.

Shopping bag and checkout

The purchase flow connects product variants, delivery details, promotional pricing and order creation.

Shopping bag

Update quantities, remove items and choose which items to check out. Availability warnings do not disclose exact stock counts. Guest carts are stored locally; customer carts have server persistence.

Delivery and payment

Select saved addresses or supply delivery details. The implemented payment choices are cash on delivery and direct bank transfer; payment status and order status are tracked separately.

Promotions

Preview automatic discounts and promo codes, show savings and totals, and access assigned offers from My Offers.

Order history

Filter orders by status, inspect items and financial details, and follow Pending, Confirmed, Processing, Shipping and Delivered stages. Eligible cancellations are handled through the order API.

Account and client care

The account area brings personal settings and purchase history together.

Profile and addresses

Edit profile information and marketing preferences. The address book supports a default delivery address and up to two additional addresses for checkout selection.

Address suggestions

The Google address integration provides suggestions and verification when API credentials and service access are configured. A map match does not itself guarantee delivery access to a property.

Reviews

Customers can submit and manage eligible product reviews. Refunds and unresolved or failed return requests block review eligibility; successfully completed exchanges can remain eligible.

Support and brand content

Our Story, contact information, shopping guidance, terms, privacy, cookies and returns pages are available. The footer includes social links and a newsletter interface. Rue provides fashion assistance when its external AI service is configured.

Submit and track a request

From Account → Orders, the customer opens the request action for the relevant item and chooses return and refund or exchange.

Request information

Select a reason and quantity, describe the issue and, for exchanges, select a valid product size and colour combination. The main account request form does not require photo upload.

Initial decision

The server verifies order ownership and prevents duplicate requests for the same order item. It returns an automatic authorisation, automatic decline or manual review decision with a reason.

Physical assessment

Initial authorisation allows the return process to begin. Maison Rue must receive and physically inspect the item before completing a refund or exchange.

Customer communication

Customers can see request decisions and progress. Post inspection approval messaging refers to processing within three working days and directs customers to support; actual fulfilment still requires operational action.

Automatic decision rules

The current policy engine applies ordered rules to each requested order item. Automatic acceptance is provisional return authorisation.

Automatic decline

More than 14 calendar days after the recorded delivery timestamp; invalid requested quantity; or a promotional item returned for fit, preference or change of mind rather than a store fault.

Manual review

Unconfirmed delivery; an unrecognised or Other reason; requested value of at least VND 10,000,000; at least three units in the request; or an unconfirmed unused, unwashed and unaltered condition for fit or preference returns.

Automatic authorisation

Within the time and risk limits: wrong item, wrong colour, wrong size sent, missing item, arrival damage or reported manufacturing defect. Eligible regular price fit or change of mind requests require condition confirmation.

Rule boundaries

Store fault reasons bypass the promotion exclusion, but not the earlier time limit. Value and quantity thresholds are evaluated per request, not across all requests in an order.

Staff and administrator portals

Both operational roles can access return handling. Administrators have additional management and reporting functions.

Staff workflow

Review operational dashboards and orders, advance supported order statuses, confirm payment, update stock at product or variant level and review inventory movement records.

Return handling

Authorise manual cases, record transit and receipt, pass or fail physical inspection with notes, then complete an eligible refund or exchange. Both staff and admin are allowed by the return API.

Inventory accounting

Return completion restores the requested quantity to the original inventory record. Exchange completion also deducts the replacement variant quantity, using database transactions and inventory movement records.

Administrative control

Manage employee accounts and active status; view orders, revenue reports and product insights; manage promotions and approvals; review and moderate customer reviews. A separate order payment refund route is admin only.

Promotions and business insight

Management tools support merchandising decisions and monitor sales performance.

Promotion management

Create and edit promotions, change status, approve campaigns and archive them. Preview or simulate promotional pricing before use, and distribute eligible codes or offers to customers.

Product and financial insight

Review revenue reporting, product insights, SKU business metrics and promotion reports. Variant pricing has an administrator restricted update endpoint.

Traceability

Order item snapshots preserve purchase time information, including promotional allocation. Inventory movements and return records connect operational decisions to affected items.

Review management

Public product reviews have customer eligibility checks and administrative moderation, linking customer feedback with actual purchase and return outcomes.

Core records and relationships

SQL Server provides the shared operational data layer for the three portals.

Identity and customer records

Users and roles → customer profile, delivery addresses, preferences, favourites and saved cart. Employee accounts support staff and admin access.

Catalogue and inventory

Products → variants, colours, sizes, prices, materials and collections. Variant stock and movement records track inventory changes.

Orders and service

Customer → orders → order items and purchase snapshots. Order items → return requests → inspection and resolution records. Reviews depend on purchase and return eligibility.

Promotions and reporting

Promotions and customer offers → pricing allocations on order items. Orders, costs, stock and SKU metrics feed reports and merchandising insights.

Integration and delivery boundaries

The website includes implemented workflows and several integration dependent services. These distinctions matter when demonstrating or deploying it.

External services

Google address suggestions require a working API key and enabled services. Rue calls an external AI API when configured and has a fallback response when no key is present.

Payment and newsletter

COD and bank transfer are the checkout choices. Recording a refund in the application should not be presented as an automatic bank payout. The newsletter interface alone does not establish a production mailing integration.

Automation scope

Return triage and Complete the Look are rule based. Physical inspection remains a human operation. Purchase history learning can be added later with suitable data.

Demonstration sequence

Browse → save a guest wishlist → sign in → choose a variant → checkout → inspect a delivered order → submit a return → handle inspection in an operational portal → inspect the customer result.

Thank you

A connected fashion shopping experience, from discovering a piece to caring for it after purchase.

Customer

Discovery, styling suggestions, saved favourites, checkout and transparent service requests.

Staff

Order fulfilment, inventory handling and physical return assessment.

Administrator

Employee management, commercial controls, reports and customer feedback oversight.